

AI Encyclopedia-English Spelling

1. Function Description

After the program starts, ask the large model what the word on the desktop might mean. After thinking, the large model will reply with the answer to the question and will check the spelling of the word, and control the robotic arm to point to the missing word.

2. Startup

After entering the Docker container, enter the following commands in the Docker terminal.

```
ros2 launch largemodel largemodel_control.launch.py text_chat_mode:=True
```

Then open a second terminal and input the following command,

```
ros2 run text_chat text_chat
```

Then input "Guess what the word on the desktop might mean?" in the text_chat terminal and press Enter; if it's the voice version, wake up the voice module and directly say to the voice module "Guess what the word on the desktop might mean?", then wait for the large model to think and reply, as shown in the figure below,

```
user input: Guess what the word on the desktop might mean?
okay@, let me think for a moment... [[INFO] [1775545695.294304341] [text_chat_node]: Decision making AI planning:Execute in accordance with the following steps, and only output one action function at a time:
1. Call 'seewhat()' to observe the environment and check the word(s) on the desktop;
2. Reply with what the word(s) on the desktop might be;
3. Call 'seewhat()' to observe the environment and check whether the word(s) on the desktop are spelled completely;
- If incomplete, execute in accordance with the following steps, and output one function at a time:
(1) Call 'seewhat()' to observe the environment and obtain the positions of the letters that can complete the spelling of the word(s);
(2) Call the 'point to(x1, y1, x2, y2)' function to point to the letters that can complete the spelling of the word(s) (where x1, y1, x2, y2 are the relevant coordinate parameters of the letters used to complete the spelling of the word(s)).
- If complete, call the 'finishtask()' function.
[INFO] [1775545700.90693758] [text_chat_node]: "action": ['seewhat()'], "response": Okay, I'm going to take a look at what's on the desktop. Let me see what words are there!
[INFO] [1775545707.877374631] [text_chat_node]: "action": ['seewhat()'], "response": Hmm, I see the letters D, u, k, and d on the desktop. They might be trying to spell 'Duck'-but it's a bit scattered! Let me check if all the letters are there and properly arranged.
[INFO] [1775545716.568234982] [text_chat_node]: "action": ['seewhat()'], "response": I see the letters D, u, k, and d again, but there's also a small text box with some characters that look like they might be part of the word. It seems like the word 'Duck' is almost complete, but it's not arranged properly. Let me check if there are any missing letters or if the spelling can be completed.
[INFO] [1775545728.158546856] [text_chat_node]: "action": ['point to(580, 380, 540, 420)'], "response": Ah, I see now! The letters D, u, k, and d are present, but the word 'Duck' is missing the letter 'c'. It looks like the 'c' is partially visible at the bottom right. Let me point to that letter so we can confirm it's there!
[INFO] [1775545761.048245232] [text_chat_node]: "action": ['finishtask()'], "response": I've found the missing letter 'c'! Now that all the letters are accounted for-D, u, c, k-it looks like the word on the desktop is 'Duck'. The spelling is complete now!
"
```

Then the robotic arm will point to the letter c,



3. Task Planning

1. Call `seewhat()` to observe the environment and check the word(s) on the desktop;
2. Reply with what the word(s) on the desktop might be;
3. Call `seewhat()` to observe the environment and check whether the word(s) on the desktop are spelled completely;
 - If incomplete, execute in accordance with the following steps, and output one function at a time:
 - (1) Call `seewhat()` to observe the environment and obtain the positions of the letters that can complete the spelling of the word(s);
 - (2) Call the `point_to(x1, y1, x2, y2)` function to point to the letters that can complete the spelling of the word(s) (where x_1, y_1, x_2, y_2 are the relevant coordinate parameters of the letters used to complete the spelling of the word(s)).
 - If complete, call the `finishtask()` function.